CALCULUS · UNIT 2: LIMITS AND CONTINUITY

Toll Road Pricing

Use limits and continuity to design fair prices for a new toll road and parking garage.

CP2

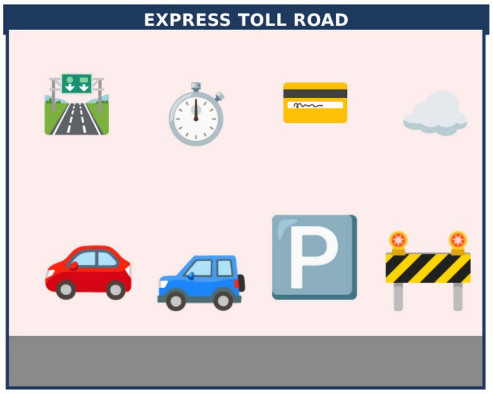
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The state highway authority is opening a new express toll road with a parking garage at the end. Drivers complain when prices jump suddenly, and the authority wants prices that are fair and predictable.

You will use limits and continuity to analyze the current prices and propose better ones.



Garage
$3 per started hour
max $20 per day

Toll
Rush hour
price changes at 7 a.m.

Learning Goals

- I can estimate limits from tables and graphs.
- I can find one-sided limits at a jump.
- I can decide whether a function is continuous at a point.

Project Timeline

When	What you will do
Day 1	Garage Prices
Day 2	Toll Plan A
Day 3	Fix the Price
Final	Pricing Proposal + reflection

Part 1: Garage Prices

Day 1 · The garage charges $3 for each hour or part of an hour.

Pricing Data Cards

Garage	Toll Plan A	Toll Plan B
$3 for each hour	T(x) = 2 + 0.1x, x < 20	T(x) = 2 + 0.1x, x < 20
or part of an hour	T(x) = 5, x ≥ 20	T(x) = k + 0.05x, x ≥ 20

Monthly Pass	Traffic	Limit Table
$40 + $0.50 per trip	6 a.m.: 900 cars/h	f(x) = (x² − 9)/(x − 3)
	8 a.m.: 2,400 cars/h	

1. Make a table of the price for 0.5, 1, 1.5, 2, and 2.5 hours.

2. Find the limit of the price as $t \to 1^-$ and as $t \to 1^+$.

3. Does lim_{t→1} P(t) exist? Explain.

   Starter: *Left limit = ___, right limit = ___*

4. Is the garage price continuous at t = 1.5? Explain.

Part 2: Toll Plan A

Day 2 · T(x) = 2 + 0.1x for x < 20 miles and T(x) = 5 for x ≥ 20.

5. Find $\lim_{x\to 20^-} T(x)$ and $\lim_{x\to 20^+} T(x)$.

6. Is Toll Plan A continuous at x = 20? Why do drivers complain?

7. Use a table to estimate $\lim_{x\to 3} \frac{x^2-9}{x-3}$.

8. Find the same limit algebraically.

Part 3: Fix the Price

Day 3 · Toll Plan B: $T(x) = 2 + 0.1x$ for $x < 20$ and $k + 0.05x$ for $x \geq 20$.

9. Find k so that Plan B is continuous at $x = 20$.

10. Find the Plan B toll for 30 miles.

11. Traffic was 900 cars/h at 6 a.m. and 2,400 at 8 a.m. Explain why traffic must have been exactly 1,500 cars/h at some time in between.

Final Product: Pricing Proposal

Your final product must include:

- Garage price table and graph
- One-sided limits at the jumps
- Continuous Plan B
- IVT traffic explanation
- Pricing proposal

Reflection

R1. What does it mean for a price function to be continuous?

R2. Why can a limit exist even if the function is not defined there?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Limits and continuity conditions are applied correctly to analyze and design fair prices. Goes beyond the minimum with a creative, realistic choice.	Limits and continuity conditions are applied correctly to analyze and design fair prices.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes graphs, limit work, continuity conditions, and a recommendation. Polished, organized, and visually engaging.	Proposal includes graphs, limit work, continuity conditions, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.